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Introduction

Cryptocurrencies, and Bitcoin in particular, has matured into a derivatives-first market where term-structure dislocations are large, persistent, and increasingly macro-relevant, transforming into a genuine institutional opportunity. For scale, COMEX Gold Futures should have roughly $149B of notional exposure outstanding[1], whereas CME reported record crypto-derivatives notional OI of $39B as of Sep-25, showcasing that institutional crypto derivatives are now a sizable opportunity that cannot be ignored. It is against this backdrop, that this project is deliberately narrow, wherein we test whether two classic term-structure strategies or "workhorses", long used in traditional commodities and rates, are widely viewed as increasingly crowded and lower edge, but can "live on" in the realm of crypto. To make a clear comparison, we hold signal design, roll rules, and cost assumptions so we can have "apples to apples" comparisons. We aim to evaluate this through two main traditional strategies:

S1: Spot-Futures Basis **S2:** Calendar Spread Mean-Reversion

This aims to answer the course question, *do these once-reliable curve trades "live on" in BTC term premia relative to the benchmarks of precious metals (gold & silver), or the same decay already caught up?*

Literature Review & Research

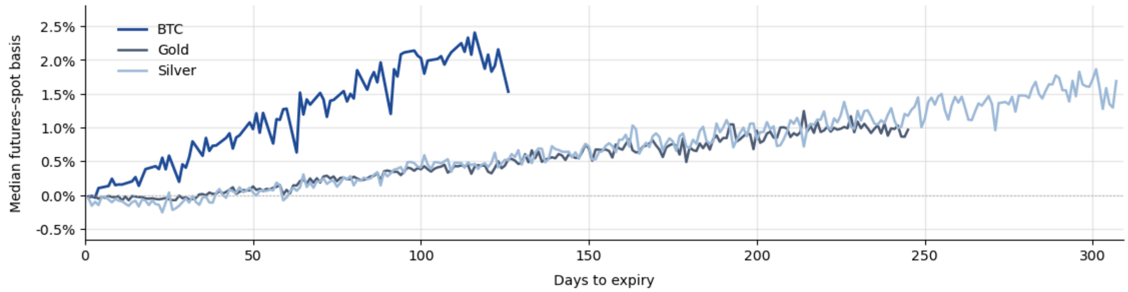
Term structure strategies exploit the notion that future curves embed a measurable carry component (financing/roll and implied yield), that can be harvested with limited spot reliance. Gorton (2004) noted that in commodities, real-life economics drive predictable backwardation/contango for commodities (EG: hedgers for WTI, driving backwardation), leading to economically meaningful premia and eventual convergence as the foundation for basis and roll-down strategies.

Resultingly, crypto is a natural testbed for these "workhorse strategies: that have decayed, as derivatives dominate the markets, and the curves are largely shaped by leveraged long demand and funding conditions. Recent evidence and practitioner-style analysis indicates that OI and basis can move together, and basis trading has become an institutional use case, especially since the post-ETF growth from the beginning of 2024 (CME Group 2024). Evidence does show that the future-spot basis remains high, which motivates the focus on two traditional strategies, that is, whether basis convergence and calendar-slope mean reversion can still deliver net-of-cost edge in BTC (Schmeling et al. 2023).

[1] COMEX Gold futures open interest and settlement prices are reported in CME Group's Daily Bulletin. The COMEX Gold futures (GC) contract unit is 100 troy ounces. Gold notional exposure is computed as open interest × futures price × contract unit (author's calculation). CME Group reported record crypto-derivatives notional open interest of $39 billion on 18 September 2025

Strategy Overview

Crypto curves tend to be in contango, like precious metals, due to collateral constraints, and market demand for effective leveraged long positions through futures. This motivates two general "workhorse" strategies, and the below image shows the opportunity gap that has yet to be capitalized to the fullest extent by the market.



Note: First four listed futures only. BTC: monthly contracts (~30 days per tenor); Gold: bi-monthly (Feb/Apr/Jun/...) (~60 days per tenor); Silver: Mar/May/Jul/Sep/Dec (~70–80 days per tenor).

The Spot-Futures basis trade (S1) involves trading the annualized basis (Future - Spot) as the "carry", with a higher basis implying a richer carry for a long-spot/short-future position. The trade is betting on convergence, with USD-notional matching, that takes no directional bets. This strategy is not purely academic, and has been implemented institutionally, for instance Millenium and Brevan Howards actively conducted such basis trades with BTC ETFs[2].

S2 is a relative value implementation that was inspired by shape-reversion term-structure strategies, trading curve shape rather than outright levels, with the near-next calendar slopes (spread between front and later), and standardizing it using a rolling half-year window. Larger values can indicate that it is unusually steep or flat relative to recent regimes, and this is implemented with strict entries and exits upon mean-reverting. This style of curve trading has been monitored (and likely traded) by Cumberland, a market marker under DRW[3]. Both strategies assume curve dislocations arise from positioning, roll mechanics and funding pressures that tend to mean-revert that enable limited despite convergence.

Data Universe & IS/OOS Framework

BTC term structure is studied using XBT (BTC/USD) spot and CME Bitcoin futures benchmarked against COMEX gold (GC) and silver (SI) futures. For each asset, the first three listed contracts are utilised, specifically, the daily futures settlement/closes. Contract notionals follows specs (BTC = 5 Bitcoins, GC = 100 troy oz, SI = 5,000 troy OZ) that S1 Positions need to account for notional-matching for delta-hedge. All data is sourced from Bloomberg. To limit overfitting key parameters are like entry/exit bands, and "margin", are set in-sample from 2018-2023 for IS (as BTC futures were introduced in 2018), and 2023-25 for OOS validation.

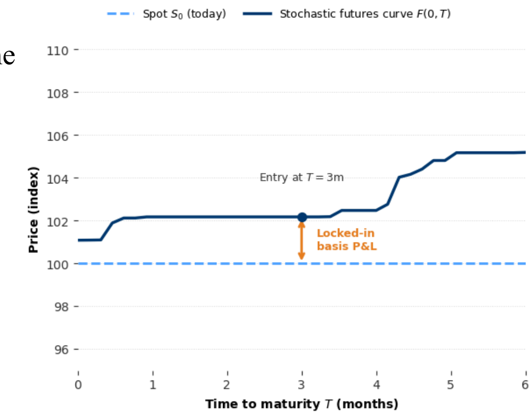
[2] Reuters, citing U.S. regulatory filings, reports both firms held U.S. spot Bitcoin ETFs and frames these positions as consistent with the "bitcoin basis trade" (buy spot bitcoin/ETFs while shorting higher-priced futures); it also notes both trimmed ETF exposure when the futures premium collapsed, supporting that large hedge funds have actively used spot–futures arbitrage in practice
[3] Cumberland explicitly lists "futures/spot basis trading" among the products it trades (alongside listed options/futures), supporting our claim that institutional crypto desks actively run basis/term-structure arbitrage strategies.

Trading Implementation

For **S1 Basis Convergence Trade** the entry signal is:

$$basis^{ann}_{k,t} = \left(\frac{F_{k,t} - S_t}{S_t}\right)\frac{365}{DTE_{k,t}}, \qquad k \in \{2M, 3M\}.$$

The trading rule that we had set was, enter the long spot / short F, when $Basis_t > \tau,\ where\ \tau$ is set as 0.05 for BTC, 0.02 (annualised basis), Exit when $Basis_t > \tau_{exit}$ or DTE < 7 (forced exit/roll). The sizing of the trade is such that the futures are chosen such that the USD notional of the post and futures are matched on 1:1 notional. This helps keep the overall position direction-neutral, so that the P&L is driven by basis decay rather than the spot direction. The Daily Mark-to-Market Pnl is $\Delta\Pi_t = N_{spot}\Delta S_t - N_{fut}\Delta F_t - costs.$

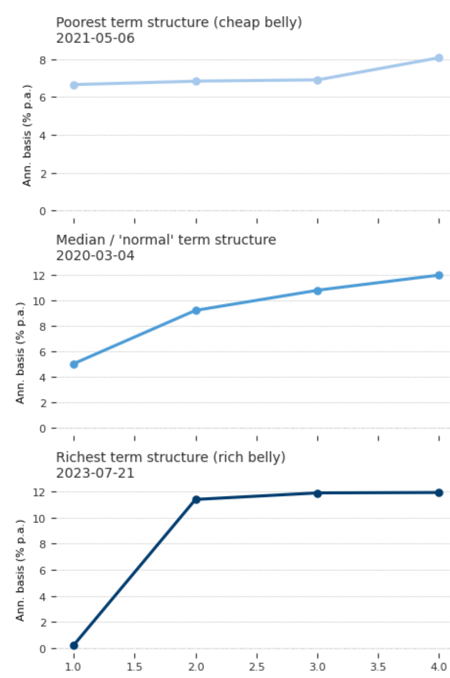


The idea is that even if convergence does hold, the basis can widen before it compresses, producing interim drawdowns that reduce Sharpe heavily, as shown in the "toy" visual on the RHS.

For **S2, calendar-spread mean-reversion,** the calendar slope and z-score, the trading signal is:

$$\mathrm{Cal}_t = \frac{F_t^{next} - F_t^{near}}{F_t^{near}} \qquad z_t = \frac{\mathrm{Cal}_t - \mu_L}{\sigma_L}$$

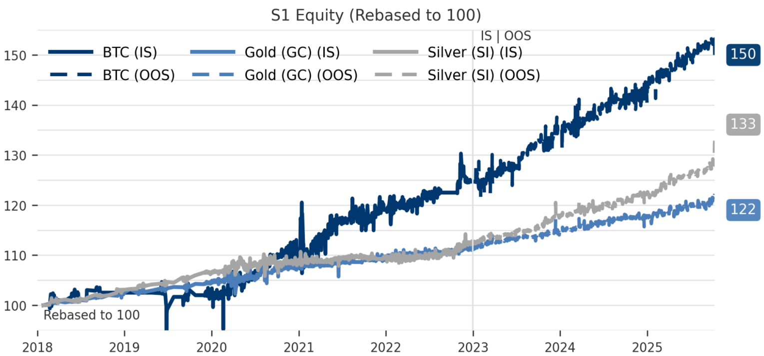
The trading rule, is that $z_t > 1.6$ $(short\ near/long\ next), z_t < -1.6\ (long\ near/short\ next)$, and exit when $|z_t| < 0.2$ or when the before the contract is forced to roll to the next pair, for example from F3 to F2. For instance, referring to the RHS for BTC term structure, for the rich belly, we will short the 2-month and long the 1-month future and hold until before expiry for the mean-reversion, but we choose to enter the richest signal for the actual trade. The Daily Mark-to-Market Pnl is $\Delta\Pi_t = N_{F1}\Delta F1_t + N_{F2}\Delta F2_t - costs.$

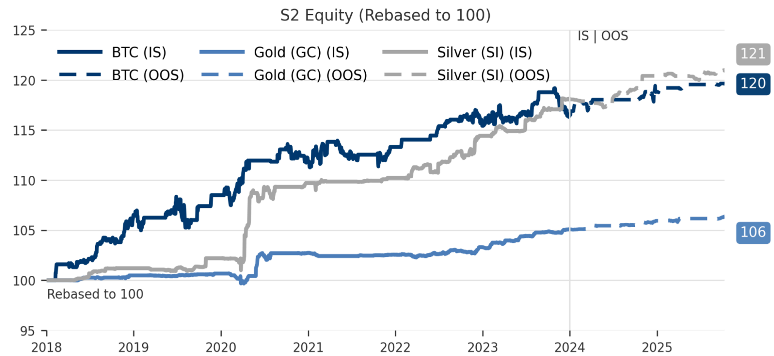


Transaction costs are modelled conservatively and symmetrically. In S1, we report results under an notional-proportional cost-model,specifically, a single bps rate applied to traded notions, whereas S2 used one tick of slippage of entry and exit, with the BTC spot incurring a fixed-basis slippage. Both are included as execution quality varies materially by venue and instrument, and a tick-based model, can understate costs under stress, while bps model can overstate in normal conditions. Portfolio construction remains unchanged, with USD-notional matched, that is S1 long-spot/short-futures, S2 has matched notionals, with positions rolled for forced closed when DTE < 3 for BTC and DTE < 7 for GC/SI given longer contract tenors.

Performance Estimate & Results

Period	Strategy	Ann. return (%)	Ann. vol (%)	Sharpe	Max DD (%)	Calmar	VaR 95% Daily	CVaR 95% Daily
2018-25	S1: BTC	6.40%	19.56%	0.415	-17.60%	0.364	-1.59%	-2.70%
2018-25	S1: GC	2.64%	6.36%	0.442	-4.70%	0.562	-0.57%	-0.95%
2018-25	S1: SI	3.76%	6.17%	0.629	-3.94%	0.955	-0.54%	-0.89%
2018-25	S2: BTC	2.246%	2.609	0.864	-2.857	0.786	-0.24	-0.385
2018-25	S2: GC	0.77%	0.558	1.377	-1.01	0.762	-0.023	-0.054
2018-25	S2: SI	2.403%	1.57	1.521	-1.172	2.05	-0.043	-0.117





Across the test period, S1 delivers by far the strongest headline returns in BTC, but with materially higher risk and drawdowns of 20%, and 18% approx. , respectively. On the other hand, the traditional GC and SI basis has far lower returns with smaller drawdowns and tighter tail risk. For S2, risk is lower across all assets, and Sharpes are higher, but returns are especially modest, with BTC performing similarly to SI, and slightly better OOS. Full IS/OOS results and VaR/CVar summary and return distribution plots in the Appendix.

Risk, Robustness & Conclusion

Although both S1 & S2 are on a fundamental level engineered to harvest term-structure premia, limited directional exposure, realised PnL, is strongly shaped by asset-specific structure. BTC term structure behaves differently from GC and SI, with the general headline performance of better or equal returns in BTC, but materially higher MTM volatility.

Market structure and regime risks exist for both strategies, where BTC's curve is tied to leverage demand and collateral conditions, such that in periods of higher risk appetite, basis remains rich, but in deleveraging episodes, the curve can flatten and invert like during the the FTX crash[4]. Refer to Appendix. The regime dependence increases dispersion in S1 and to a lesser extent, S2. BTC is notably more prone to liquidation and gap risk[5], wherein stressed order books and forced deleveraging can widen basis or calendar slopes, and therefore large interim drawdowns, even when S1 is notionally delta-neutral and S2 is a calendar spread, in

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[4] During severe deleveraging, BTC term structure can flip from persistent contango into sharp flattening/backwardation. For example, BIS evidence shows CME (futures–spot basis) ranged from below −50% during the FTX collapse (Nov 2022) to strongly positive levels in calmer regimes, consistent with regime-dependent curve behaviour

[5] BTC, and in fact any crypto is more exposed to liquidation from margin class and liquidity gaps than GC/SI, as the latter two are far less volatilte: forced deleveraging can reduce order-book depth and widen spreads, amplifying short-horizon basis/slope moves and increasing outcome dispersions.

practice, increasing margin-call risk. But BTC has its own idiosyncratic risks that are not present in traditional commodities, with custody/key management risks being an example, with custodians like Coinbase suffering from multiple attacks in recent years. This can be mitigated from spot exposures through ETFs like BlackRock's IBIT. There are other risks like a theoretical 51% attack (majority hashpower), that can enable transaction censorship. While prohibitively expensive for BTC (like buying 51% ownership), it remains a non-zero systemic risk that does not exist for GC/SI.

Looking forward, potential improvements mainly involve making the trade more asset-risk aware, that can be implemented would be Volatility targeting or OU-processes. Volatility targeting on the basis/spread, where the same entry/exit logic is used, by exposure is scaled by recent realised volatility, such that positions automatically de-risk, when the underlying becomes unstable, with:

$$w_t = min(w_{max}, \frac{\sigma^*}{\widehat{\sigma}_t + \epsilon}),\ Position_t = w_t * BasePosition.$$

This directly targets the noted failure in BTC with the larger MTM swings inflating volatility and depressing Sharpe, despite convergence. Another alternative suggested by Dr Baker from the CU Statistics Department would be a Expiry-anchored Ornstein-Uhlenbeck/ bridge scoring [6], where the basis b_t, can be modelled as a mean-reverting process that is explicitly conditioned to converge near expiry, with a bridge style assumption of

$$db_t = -\ kb_t dt\ +\ \sigma dW_t,\ where\ b_T = 0\ at\ T.$$

The idea is that this can the potential computation of a a predictive distribution for the basis over holding horizon *h*,

$$b_{t+h} | (b_t, b_T = 0) \sim N(\mu_{t,h}, \sigma_{t,h}^2) \rightarrow Score_t = \frac{b_t - \mu_{t,h}}{\sigma_{t,h}}$$

And a probability filter can be applied such as $p_t = P(b_{t+h} > 0\ |\ b_t, b_T = 0)$, which measures the strength of expected convergence. Operationally, this allows for trade entries if the model implies strong expected tightening, or larger $b_t - \mu_{t,h}$ values, and lower implied dispersions, like smaller $\sigma_{t,h}$. This directly targets the crux of the main issue in S1, where convergence does happen, but the path can still generate margin stresses and heighten volatilities.

BTC remains a more attractive but higher-risk venue for the classic term-structure "workhorse trades", with larger potential carry, but interim MTM volatility and operational tail risks are materially higher than GC/SI, which restricts Sharpe and increases drawdowns. From an allocation perspective, S1 might be good, considering decent returns and low correlations with major assets. The forward steps are to integrate more mathematically rigorous models, like volatility targeting, and DTE-aware sizing predicated on OU-Bridge convergence perspectives, using the core strategies as a foundation.

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[6] Bridge-style modelling motivation (and "convergence can still hurt"): Liu & Longstaff (2004) model arbitrage value as a bridge-type process that converges at a fixed horizon, but show that with collateral/margin constraints it can be optimal to under-invest and that strategies can experience large interim losses before convergence. This means that rather than the 1:1 notional like the basis trade done above, it can be a better idea to utilise this trade, while maintaining some margin above the maintenance margin, such that this trade will have levered returns above the RFR.

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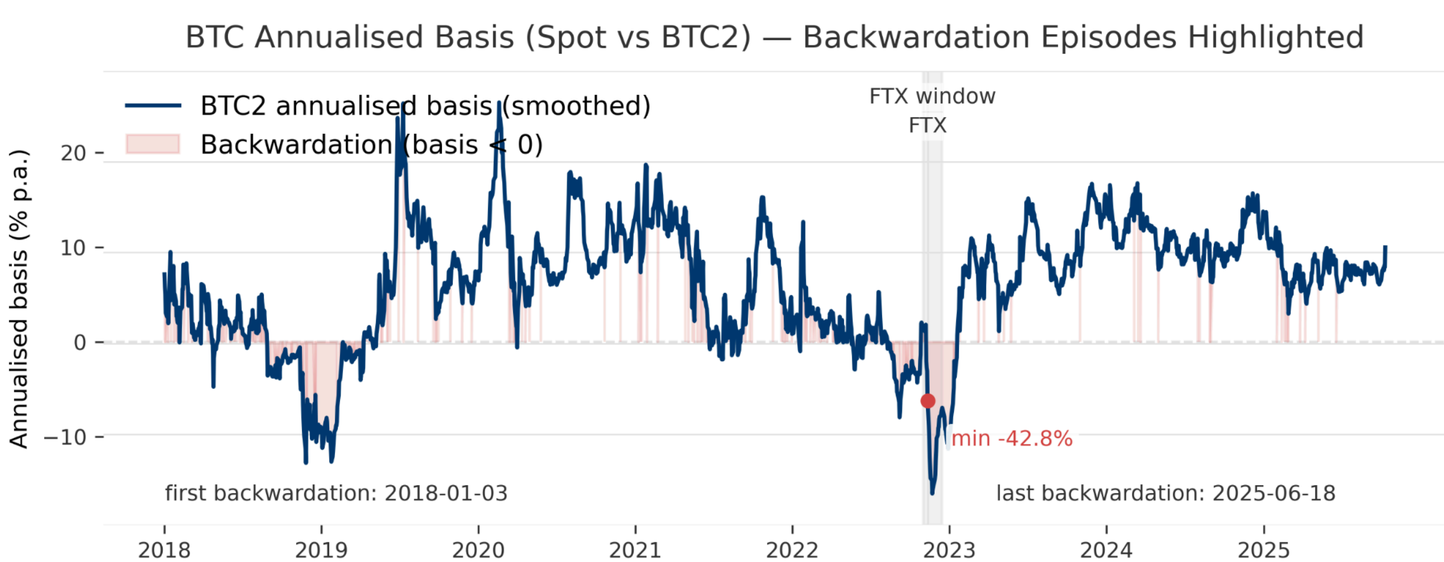
Appendix

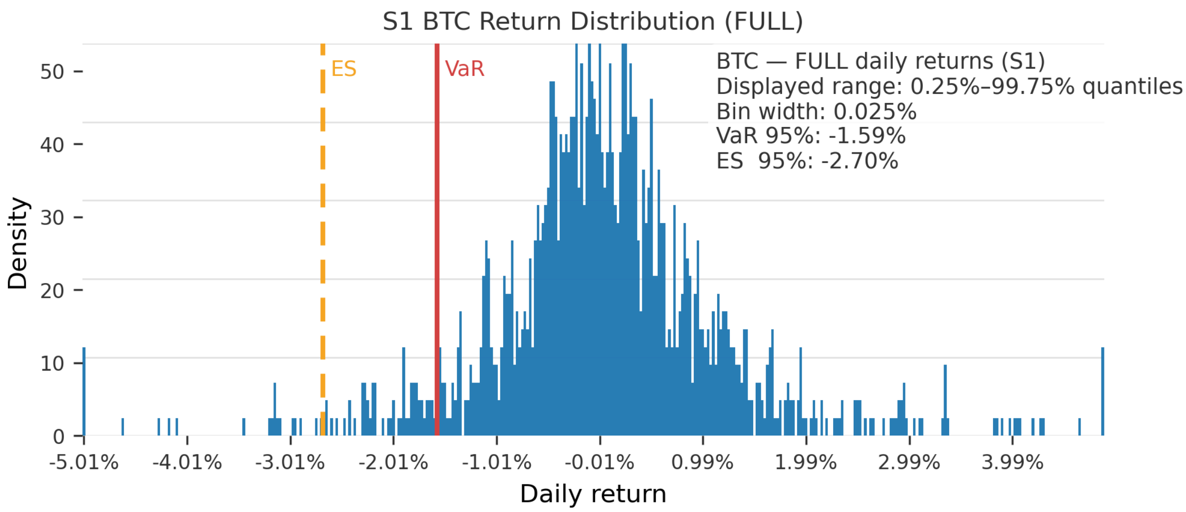
BTC Historical Basis

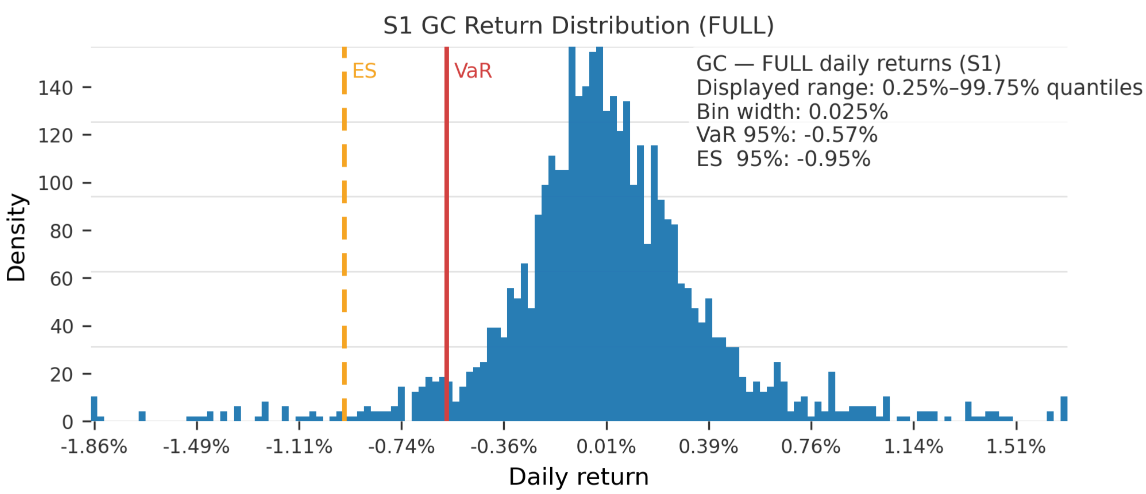


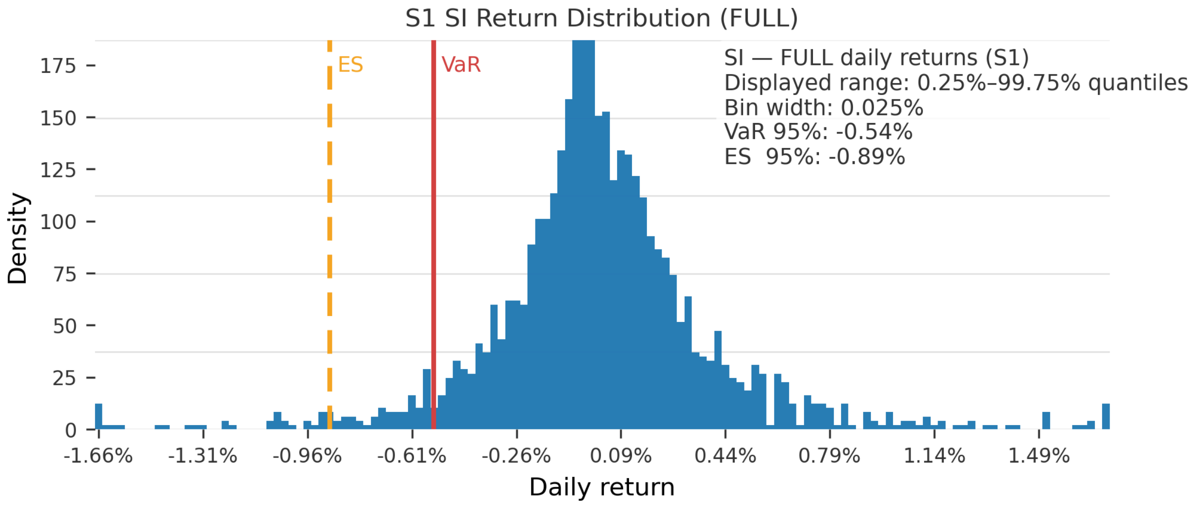
S1: Summary Statistics

Period	Asset	Ann. return	Ann. vol	Sharpe	Max DD	Calmar	VaR 95% (daily)	CVaR 95% (daily)
FULL	BTC	6.40%	19.56%	0.415	-17.60%	0.364	-1.59%	-2.70%
IS	BTC	6.04%	21.75%	0.379	-17.60%	0.343	-1.66%	-2.90%
OOS	BTC	6.90%	16.15%	0.494	-5.70%	1.211	-1.53%	-2.41%
FULL	GC	2.64%	6.36%	0.442	-4.70%	0.562	-0.57%	-0.95%
IS	GC	2.25%	6.36%	0.381	-4.70%	0.477	-0.52%	-0.96%
OOS	GC	3.45%	6.35%	0.566	-2.74%	1.258	-0.60%	-0.91%
FULL	SI	3.76%	6.17%	0.629	-3.94%	0.955	-0.54%	-0.89%
IS	SI	2.46%	6.26%	0.42	-3.94%	0.625	-0.57%	-0.91%
OOS	SI	6.25%	6.03%	1.036	-2.71%	2.304	-0.50%	-0.84%

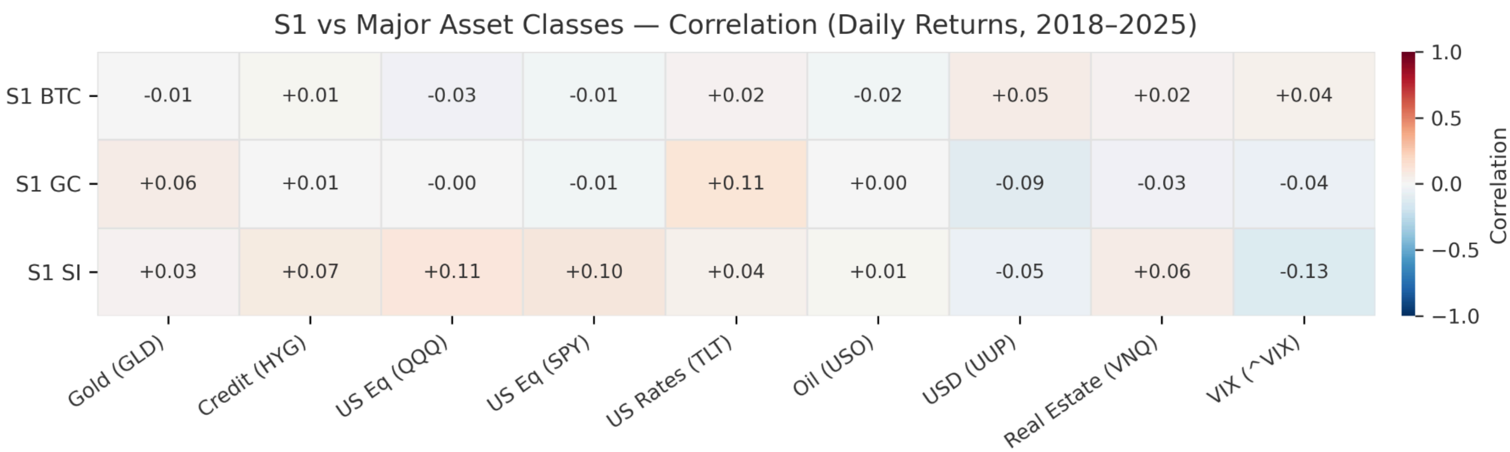
S1: Return Distributions







S1 Correlation Matrix

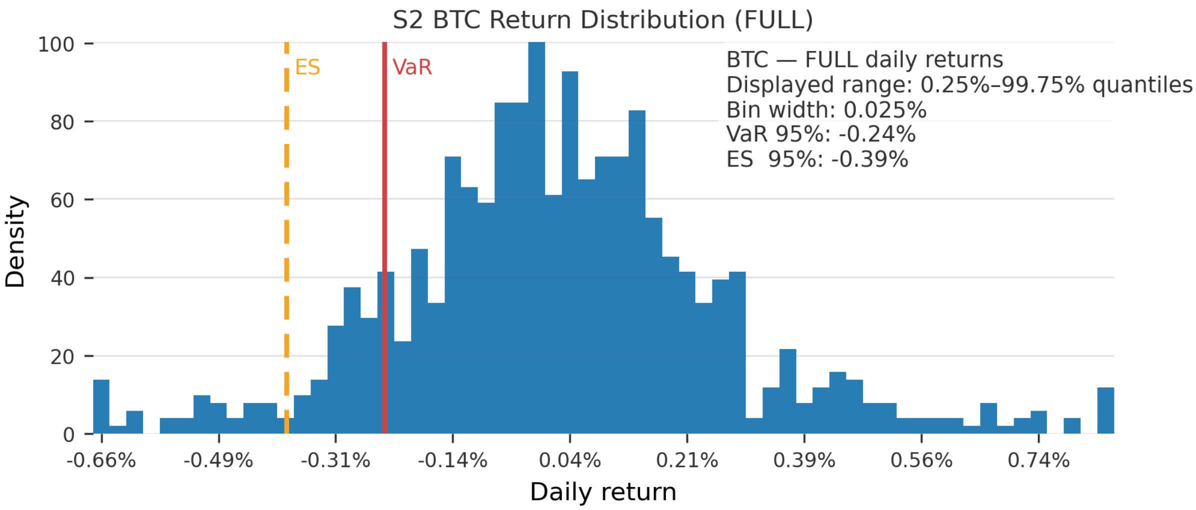


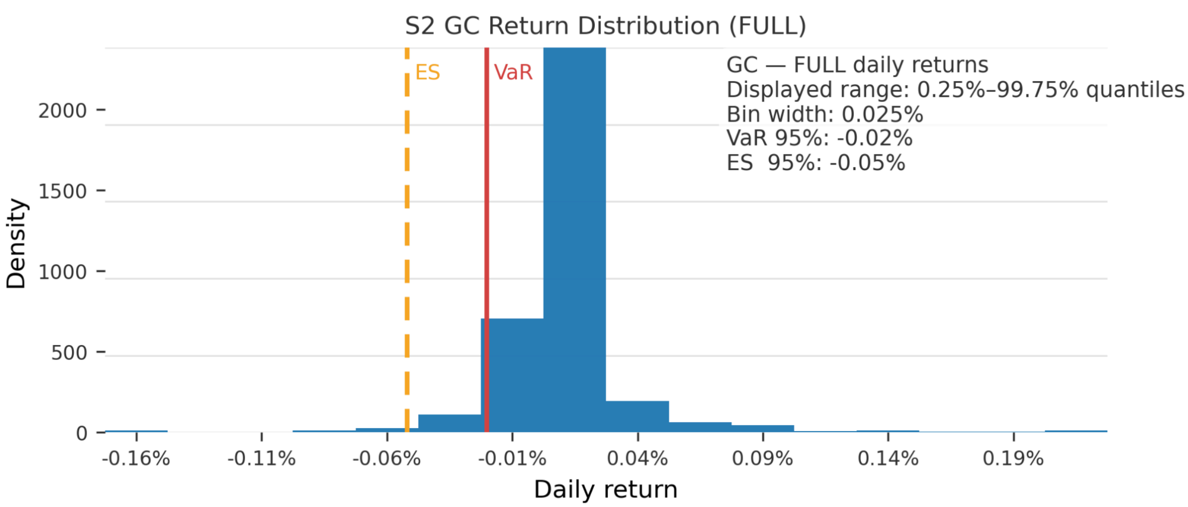
	Gold (GLD)	Credit (HYG)	US Eq (QQQ)	US Eq (SPY)	US Rates (TLT)	Oil (USO)	USD (UUP)	Real Estate (VNQ)	VIX (^VIX)
S1 BTC	-0.01	+0.01	-0.03	-0.01	+0.02	-0.02	+0.05	+0.02	+0.04
S1 GC	+0.06	+0.01	-0.00	-0.01	+0.11	+0.00	-0.09	-0.03	-0.04
S1 SI	+0.03	+0.07	+0.11	+0.10	+0.04	+0.01	-0.05	+0.06	-0.13

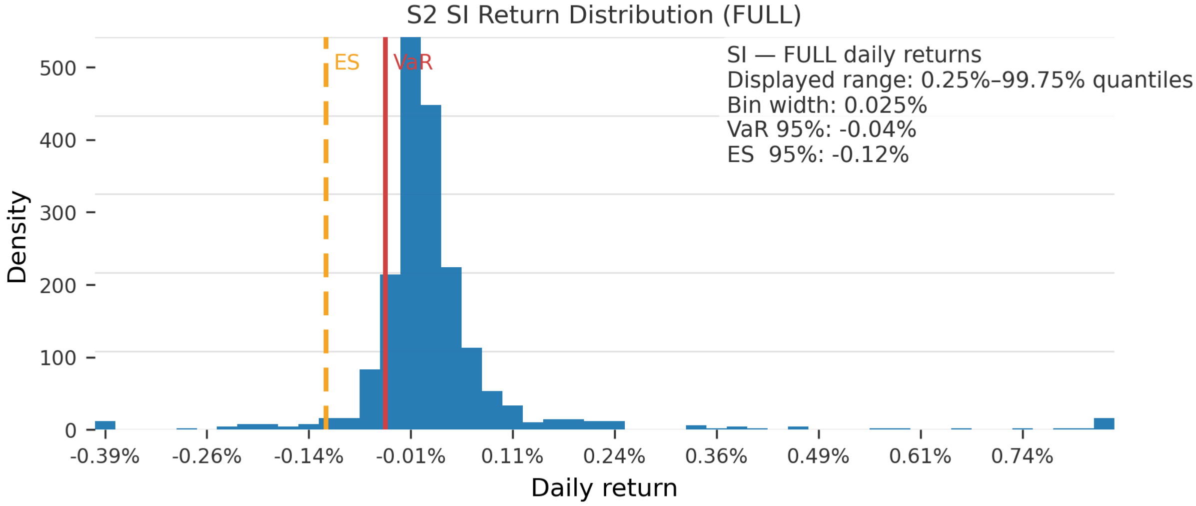
S2: Summary Statistics

Period	Asset	Ann. return	Ann. vol	Sharpe	Max DD	Calmar	VaR 95% (daily)	CVaR 95% (daily)
IS	BTC	2.49	2.899	0.862	-2.857	0.871	-0.265	-0.418
OOS	BTC	1.435	1.19	1.201	-0.908	1.58	-0.05	-0.147
FULL	BTC	2.246	2.609	0.864	-2.857	0.786	-0.24	-0.385
IS	GC	0.804	0.616	1.302	-1.01	0.796	-0.023	-0.06
OOS	GC	0.65	0.281	2.3	-0.191	3.394	-0.02	-0.036
FULL	GC	0.77	0.558	1.377	-1.01	0.762	-0.023	-0.054
IS	SI	2.728	1.71	1.582	-1.172	2.327	-0.043	-0.132
OOS	SI	1.311	0.948	1.376	-0.734	1.786	-0.045	-0.064
FULL	SI	2.403	1.57	1.52	-1.172	2.05	-0.043	-0.117

S2: Return Distributions







S2 Correlation Matrix